Summary in Graph

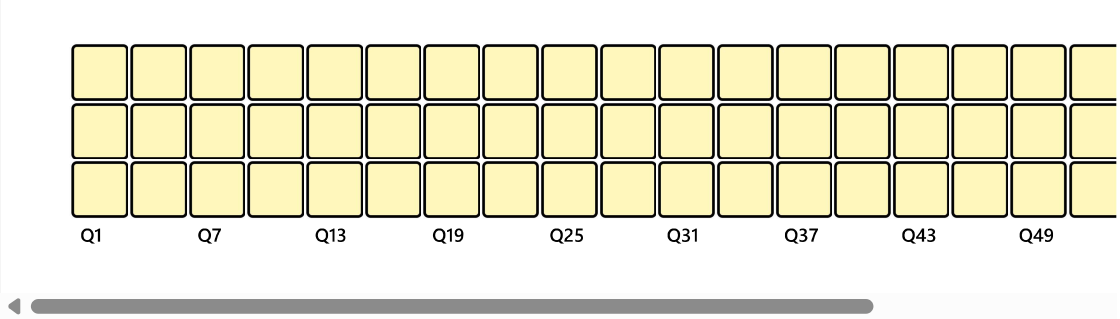
Exam Summary (GO Classes DA 2026 | All India Mock Test 6)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**65** (30 + 35)
Total Marks:	**100** (30 + 70)
Exam Duration:	**180 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (25) out of 249 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Prof. Murphy likes to let her students choose who their partners will be; however, no pair of students may work together for more than seven class periods in a row.

Alice and Bob have worked together for seven class periods in a row. Calvin and Denny have worked together for three class periods in a row. Calvin does not want to work with Alice.

Who should be assigned to work with Bob?

A. Calvin

B. Alice

C. Denny

D. None

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Four friends were sharing a pizza. They decided that the oldest friend will get an extra piece of pizza.

Bahu is two months older than Kattappa. Kattappa, in turn, is three months younger than Bhalla. Devsena is one month older than Kattappa.

Who should get the extra piece of pizza?

A. Bahu

B. Devsena

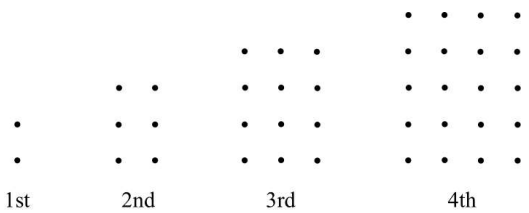
C. Bhalla

D. Kattappa

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 09sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

An *oblong number* is the number of dots in a rectangular grid with one more row than column. The first four oblong numbers are $2, 6, 12$, and 20, and are represented below:



What is the 7th oblong number?

A. 42

B. 49

C. 56

D. 64

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Given that $f(y) = \frac{|y|}{y}$, and q is non-zero real number, the value of $|f(q) - f(-q)|$ is

A. 0

B. -1

C. 1

D. 2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Select the pair that best expresses a relationship similar to that expressed in the pair:

water: pipe::

A. cart: road

B. electricity: wire

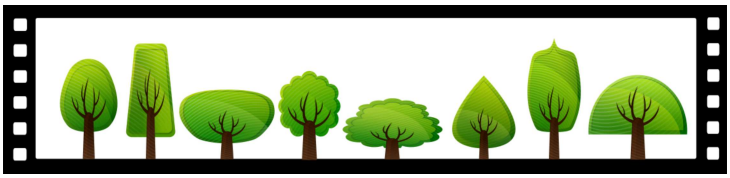
C. sea: beach

D. music: instrument

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

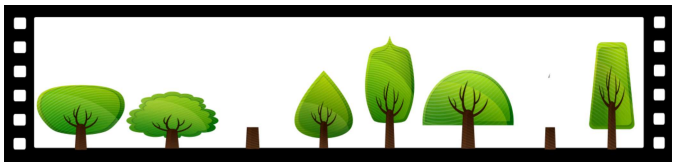
Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Dai has a camera with a *panorama mode*, which can take one continuous photo while it moves horizontally. Dai stood in the middle of a circle of eight trees, and took the following photo in panorama mode while rotating 360°.

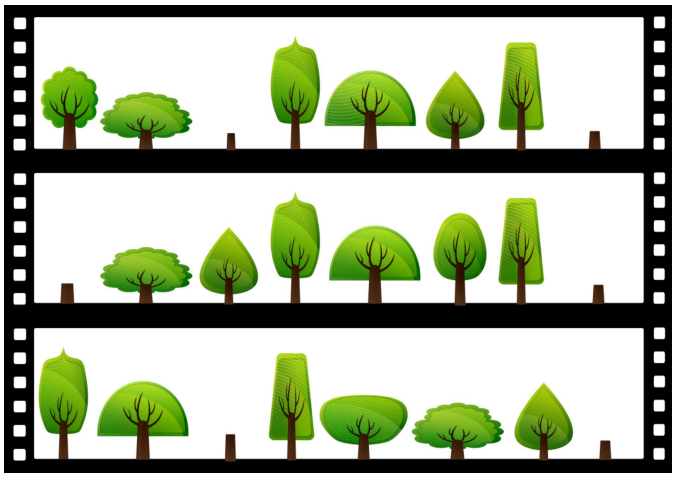


After a few days, Dai returned to the same location and took a second photo moving in the same direction, but starting from a different tree. She saw that two of the trees had been cut down.

Which of the following is Dai's second photo?

A. 

B.



C.

D.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Suppose that $N = 3x + 4y + 5z$, where x equals 1 or -1, and y equals 1 or -1, and z equals 1 or -1. How many of the following statements are true?

- N can equal 0.
- N is always odd.
- N cannot equal 4.
- N is always even.

A. 0

B. 1

C. 2

D. 3

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

For how many positive integers x is $(x-2)(x-4)(x-6)\cdots(x-2016)(x-2018) \leq 0$?

(The product on the left side of the inequality consists of 1009 factors of the form $x - 2k$ for integers k with $1 \leq k \leq 1009$).

A. 1009

B. 1010

C. 1514

D. 1515

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A and B move clockwise around a circle, starting from a common point O. A takes 9 minutes to complete a round but restarts after a delay of 1 minute. B takes 13 minutes to complete the round but restarts after a delay of 2 minutes.

How many minutes after they began would they meet again at O?

A. 30

B. 29

C. 31

D. 28

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

There are two examinations, A and B, in a subject which are evaluated out of 30 and 70 marks, respectively. In order to pass the course, the student has to get at least 40% in total and at least 40% in B. The following are the marks of the students S_1 to S_4.

Students	A	B
S_1	12	28
S_2	10	29
S_3	16	27
S_4	05	29

The only student/s to have passed is/are

A. S_1, S_3

B. S_1, S_2, S_4

C. S_1, S_2

D. S_1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a $\mathrm{n} \times \mathrm{n}$ complex matrix whose characteristic polynomial is given by
$f(t) = t^n + c_{n-1}t^{n-1} + \ldots\ldots + c_1 t + c_0$.
Then

A. $\det(A) = c_{n-1}$

B. $\det(A) = c_0$

C. $\det(A) = (-1)^n c_{n-1}$

D. $\det(A) = (-1)^n c_0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

The set $V = \{(x, y) \in R^2 : xy \geq 0\}$ is

A. A vector subspace of R^2

B. Not a vector subspace of R^2 since every element does not have an inverse in V

C. Not a vector subspace of R^2 since it is not closed under scalar multiplication

D. Not a vector subspace of R^2 since it is not closed under vector addition

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a $n \times n$ positive semi-definite matrix with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, and with α as the maximum diagonal entry. We can find a vector x such that $x^t x = 1$, where t denotes the transpose, and

A. $x^t A x > \lambda_1$

B. $x^t A x < \lambda_n$

C. $\lambda_n \leq x^t A x \leq \lambda_1$

D. $x^t A x > n\alpha$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following algorithm:

```
Alg(n):
  for i = 1 to n:
    for j = 1 to 2^i:
      print(j)
```

What is the asymptotic runtime of `Alg(n)`?

A. $\Theta(n \cdot 2^n)$

B. $\Theta(2^{n^2})$

C. $\Theta(2^n)$

D. $\Theta(n^2)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let M be a 3×2 real matrix having a singular value decomposition

$$M = USV^T,$$

where $S = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$, U is a 3×3 orthogonal matrix and V is a 2×2 orthogonal matrix.

Which of the following statements is/are *true*?

A. The rank of the matrix M is 1.

B. The trace of the matrix M^TM is 4.

C. The largest eigenvalue of the matrix M^TM is 3.

D. The nullity of the matrix M is 1.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $m > 1$ and $n > 1$ be integers. Let A be an $m \times n$ matrix such that for some $m \times 1$ matrix b_1, the equation $Ax = b_1$ has infinitely many solutions. Let b_2 denote an $m \times 1$ matrix different from b_1. Then $Ax = b_2$ has

A. infinitely many solutions for some b_2.

B. a unique solution for some b_2.

C. no solution for some b_2.

D. finitely many solutions for some b_2.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f : R^2 \to R$ be a smooth function with positive definite Hessian at every point.
Let $(a, b) \in R^2$ be a critical point of f.
Then

A. f has a global minimum at (a, b)

B. f has a local, but not a global minimum at (a, b)

C. f has a local, but not a global maximum at (a, b)

D. f has a global maximum at (a, b)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let f be the function defined by $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$ for all x such that $-1 \leq x \leq 1$. Then $f'(x) =$

A. $\frac{1}{1-x}$

B. $\frac{x}{1-x}$

C. $\frac{1}{1+x}$

D. $\frac{x}{1+x}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X \sim \text{Binomial}(n, p)$, where $0 < p < 1$. Define a new random variable

$$Y = n - X.$$

Which of the following is the distribution of Y?

A. $\text{Binomial}(n, p)$
B. $\text{Binomial}(n, 1-p)$
C. $\text{Poisson}(np)$
D. $\text{Poisson}(n(1-p))$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Let X be an arbitrary random variable that takes values in the set

$$\{0, 1, 2, \ldots, 10\}.$$

What is the maximum possible value of $\text{Var}(X)$?

Your Answer: | Correct Answer: 25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a discrete random variable with probability mass function

$$P(X = n) = 2^{-n}, \quad n = 1, 2, 3, \ldots$$

For a fixed positive integer y, the value of the probability $P(X \geq 2y)$ is

A. 2^{-2y}

B. $2^{-(2y-1)}$

C. $2^{-(y-1)}$

D. 2^{-y}

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Probability

The probability density function of the random vector (X, Y) is given by

$$f_{X,Y}(x,y) = \begin{cases} c, & 0 < x < y < 1 \\ 0, & \text{otherwise} . \end{cases}$$

Then the value of c is equal to...

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A company records heights of all employees. Let X and Y denote the errors in the average height of male and female employees respectively. Assume that $X \sim N(0,4)$ and $Y \sim N(0,9)$ and they are independent. Then the distribution of $Z = (X+Y)/2$ is

A. $N(0, 6.5)$

B. $N(0, 3.25)$

C. $N(0, 2)$

D. $N(0, 1)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Assume $n \neq d$. Suppose vectors $x_1, x_2, \ldots, x_n \in \mathbb{R}^d$ span $\mathbb{R}^d$.
What is the rank of the matrix

$$\sum_{i=1}^{n} x_i x_i^\top ?$$

A. n

B. $\min(n, d)$

C. d

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

After applying Principal Component Analysis (PCA) to a dataset, the eigenvalues of the covariance matrix are obtained as

$$\lambda_1 = 10, \lambda_2 = 6, \lambda_3 = 3, \lambda_4 = 1$$

If the data is projected onto only the first principal component, what fraction of the total variance is retained?

A. 0.3

B. 0.4

C. 0.5

D. 0.6

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a linear Support Vector Machine (SVM) with decision boundary

$$\mathbf{w}^\top \mathbf{x} + w_0 = 0.$$

Let $\mathbf{x}_p$ be the orthogonal projection of a point $\mathbf{x}$ onto the separating hyperplane. Suppose r denotes the signed geometric distance from $\mathbf{x}$ to the hyperplane, measured in the direction of the normal vector $\mathbf{w}$.

Which of the following expressions for $\mathbf{x}$ is correct?

A. $\mathbf{x} = \mathbf{x}_p + r\,\mathbf{w}$

B. $\mathbf{x} = \mathbf{x}_p + r\,\dfrac{\mathbf{w}}{\|\mathbf{w}\|}$

C. $\mathbf{x} = \mathbf{x}_p + r\,\dfrac{\mathbf{w}}{\|\mathbf{w}\|^2}$

D. $\mathbf{x} = \mathbf{x}_p - \dfrac{r}{\|\mathbf{w}\|}\,\mathbf{w}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Assume that you have given linearly separable data for a classification task. You are using support vector machines. More precisely, you are optimizing

$$\text{argmin}_{\mathbf{w}} \sum_{n=1}^{N} \left[1 - y_n \mathbf{x}_n^\top \mathbf{w}\right]_+ + \frac{\lambda}{2}\|\mathbf{w}\|^2,$$

with a very large value of λ. Recall: $[z]_+ = \max\{0, z\}$. Assume that $(\tilde{\mathbf{x}}_n, \tilde{y}_n)$ is an essential support vector, i.e., a data point that lies exactly on the margin. What is the most likely outcome if you remove this support vector from the data and rerun the algorithm with the remaining $N - 1$ data points?

A. I will get exactly the same outcome and the same margin.

B. The optimal vector $\mathbf{w}^*$ will change and the margin will decrease.

C. The optimal vector $\mathbf{w}^*$ will change and the margin will increase.

D. The algorithm will not work since the data is missing an essential support vector.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following *pseudocode* that processes an integer array $V[0 \ldots n-1]$, where $V[i]$ denotes the number of votes received by option i.

```
expandVotes(V):
  L ← empty list
  for i = 0 to n−1
    add i to L exactly V[i] times
  cursor ← 0
```

After executing `expandVotes(V)`, the following operations allow access to the elements of `L` in order:

```
hasNext():
  return (cursor < |L|)

next():
  value ← L[cursor]
  cursor ← cursor + 1
  return value
```

If $V = [0,\ 2,\ 1,\ 0,\ 1,\ 0]$, which sequence is produced by repeated calls to `next()` until `hasNext()` becomes false?

A. $1,\ 1,\ 2,\ 4$

B. $1,\ 2,\ 4$

C. $0,\ 1,\ 1,\ 2,\ 4,\ 5$

D. $2,\ 2,\ 1,\ 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

The *Single-Source Shortest Path Tree (SPST)* algorithm is executed on a weighted graph G with source vertex 0. After the execution, the distance array d and the parent/predecessor array p are obtained as follows:

Index	0	1	2	3	4	5	6	7
$d[\,]$	0	10	13	14	11	16	18	15
$p[\,]$	−1	0	4	2	1	2	7	3

Using the above information, determine the number of edges in the shortest path from vertex 0 to vertex 6.

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

You are given a hash table of size M that uses linear probing for collision resolution. The table is half-full, i.e., exactly $M/2$ cells are occupied.

The occupied and empty cells strictly alternate throughout the table. The cost of inserting an item is defined as the number of probes required to find an empty cell. Assume that the initial hash position of a key is uniformly distributed over the table.

What is the average time complexity to insert a new item?

A. $\Theta(M)$

B. $\Theta(\log M)$

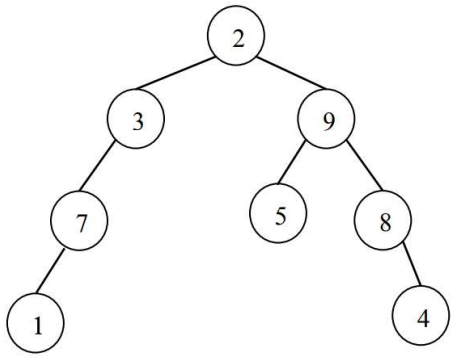
C. $\Theta(1)$

D. $\Theta(\sqrt{M})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following binary tree (the tree is *not* a Binary Search Tree):



Which of the following sequences represents the correct *preorder traversal* of the given tree?

A. $2, 3, 7, 1, 9, 5, 8, 4$

B. $2, 9, 5, 8, 4, 3, 7, 1$

C. $1, 7, 3, 2, 5, 9, 4, 8$

D. $3, 7, 1, 2, 5, 9, 8, 4$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following function that operates on a singly linked list.

Initial linked list: $\text{head} \rightarrow 1 \rightarrow 2 \rightarrow \text{NULL}$

```
def ultimateFun(head):
  if head is None or head.next is None:
    return head

  temp = head.next
  head.next = None
  temp.next = head
  head = temp

  return head
```

After the function `ultimateFun` is executed, what will be the resulting linked list?

A. $\text{head} \rightarrow 1 \rightarrow 2 \rightarrow \text{NULL}$

B. $\text{head} \rightarrow 2 \rightarrow 1 \rightarrow \text{NULL}$

C. $\text{head} \rightarrow 1 \rightarrow \text{NULL}$

D. $\text{head} \rightarrow 2 \rightarrow \text{NULL}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider a relation R with five attributes: A, B, C, D, and E. R has at least one non-trivial functional dependency. Attribute C never appears on the right-hand side of any non-trivial functional dependency applying to R. If R has exactly one candidate key, how many different possibilities are there for what this candidate key for R can be?

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider a relation schema R with attribute set A and a prescribed set of functional dependencies F over A. Which of the following statements is true?

A. There must exist a decomposition of R into $BCNF$ that is dependency preserving.

B. There must exist a decomposition of R into $3NF$ that is dependency preserving, but not necessarily a decomposition into $BCNF$ that is dependency preserving.

C. It is not guaranteed that there exists a decomposition of R into $3NF$ that is dependency preserving.

D. It is not guaranteed that there exists a decomposition of R into $2NF$ that is dependency preserving.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Let P be a compound proposition over 4 propositional variables : a, b, c, d. We know that for a compound proposition over n propositional variables, we have 2^n rows in the truth table. Every row of the truth table of P is called an "Interpretation" of P. A row in the truth table of P is called "model" iff P is true for that row. Let P be the sentence $(a \wedge b) \vee (b \wedge c)$ How many models are there for P?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A and B be $n \times n$ matrices with real entries such that

$$AB = 0.$$

Which of the following statements are *necessarily TRUE*?

(More than one option may be correct.)

A. The column space of B is contained in the null space of A.

B. $\text{rank}(B) \leq \text{nullity}(A)$

C. $\text{rank}(A) + \text{rank}(B) \leq n$

D. $\text{nullity}(A + B) \geq \text{nullity}(A)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $M = \begin{pmatrix} 1 & a & b \\ 0 & 2 & c \\ 0 & 0 & 1 \end{pmatrix}, a, b, c \in R.$

Then, M is diagonalizable, if and only if

A. $a = bc$

B. $b = ac$

C. $c = ab$

D. $a = b = c$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the matrix $A = I_9 - 2u^T u$ with $u = \frac{1}{3}[1, 1, 1, 1, 1, 1, 1, 1, 1]$, where I_9 is the 9×9 identity matrix and u^T is the transpose of u. If λ and μ are two distinct eigenvalues of A, then $|\lambda - \mu| =$ ____ .

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let R be an $n \times n$ nonsingular matrix. Let P and Q be two $n \times n$ matrices such that $Q = R^{-1}PR$. If x is an eigenvector of P corresponding to a nonzero eigenvalue λ of P, then

A. Rx is an eigenvector of Q corresponding to the eigenvalue λ of Q

B. Rx is an eigenvector of Q corresponding to the eigenvalue $\frac{1}{\lambda}$ of Q

C. $R^{-1}x$ is an eigenvector of Q corresponding to the eigenvalue $\frac{1}{\lambda}$ of Q

D. $R^{-1}x$ is an eigenvector of Q corresponding to the eigenvalue λ of Q

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0, \lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors of the matrix P corresponding to the eigenvalues λ_1, λ_2 and λ_3, respectively. Then the entry in the first row and the third column of P is

A. 0

B. 1

C. -1

D. 2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let $f : (a, b) \to \mathbb{R}$ be a differentiable function on (a, b).

Which of the following statements is/are true?

A. $f'(x) > 0$ for all $x \in (a, b)$ implies that f is increasing on (a, b).

B. f is increasing on (a, b) implies that $f'(x) > 0$ for all $x \in (a, b)$.

C. If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then there exists a right neighbourhood $(x_0, x_0 + h)$ of x_0, with $h > 0$, such that $f(x) > f(x_0) \quad \text{for all } x \in (x_0, x_0 + h)$.

D. If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then f has a local minimum at x_0.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If a and b are positive numbers, then $\lim_{x \to 0} \left(1 + \frac{x}{a}\right)^{b/x} =$

A. e^{ab}

B. $e^{b/a}$

C. $e^{a/b}$

D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If the function f is defined by $f(x) = \frac{\log x}{x^2 - 1}$ for all positive numbers x except $x = 1$, and if $f(1) = a$, for what value of a is f continuous at $x = 1$?

A. 0

B. $\frac{1}{2}$

C. 1

D. There is no such value.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let $\Omega = (0, 1]$ be the sample space and let $P(\cdot)$ be a probability function defined by

$$P((0, x]) = \begin{cases} \frac{x}{2} & \text{if } 0 \leq x < \frac{1}{2} \\ x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

Then $P\left(\left\{\frac{1}{2}\right\}\right)$ is equal to ____

Your Answer: | Correct Answer: 0.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X_1, X_2, X_3, \ldots$ be a sequence of i.i.d. random variables with mean 1.

Let N be a positive integer–valued random variable such that

$$P(N = k) = \frac{1}{2^k}, \qquad k = 1, 2, 3, \ldots$$

Assume that N is independent of the X_i's. Then the value of

$$\mathbb{E}(X_1 + X_2 + \cdots + X_N)$$

is equal to

A. 1

B. 2

C. 4

D. ∞

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let (X_1, X_2, X_3) be a trivariate normal random vector with mean vector

$$(-3,\ 1,\ 4)$$

and variance–covariance matrix

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & -3 \\ 0 & -3 & 4 \end{pmatrix}.$$

Which of the following statements are true? (Multiple selections may be correct.)

A. X_2 and X_3 are independent.

B. $X_1 + X_3$ and X_2 are independent.

C. X_1 and X_2 are independent.

D. $\frac{1}{2}(X_2 + X_3)$ and X_1 are independent.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let E_1 and E_2 be independent events such that

$$P(E_1) = \frac{1}{4}, \quad P(E_2 \mid E_1) = \frac{1}{2}, \quad P(E_1 \mid E_2) = \frac{1}{4}.$$

Define random variables X and Y by

$$X = \begin{cases} 1, & \text{if } E_1 \text{ occurs,} \\ 0, & \text{if } E_1 \text{ does not occur,} \end{cases} \qquad Y = \begin{cases} 1, & \text{if } E_2 \text{ occurs,} \\ 0, & \text{if } E_2 \text{ does not occur.} \end{cases}$$

Which of the following statements is/are correct?

A. X is uniformly distributed on the set $\{0, 1\}$

B. X and Y are identically distributed

C. $P(X^2 + Y^2 = 1) = \frac{1}{2}$

D. $P(XY = X^2Y^2) = 1$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X and Y be jointly distributed random variables.

The conditional distribution of Y, given $X = x$, is uniform on the interval $(x - 1,\ x + 1)$.

Suppose $E(X) = 1$.

The mean of the random variable Y is

A. $1/2$

B. 1

C. $3/2$

D. 2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the objects $\{1, 2, 3, 4\}$ with the distance matrix

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 11 & 5 \\ 1 & 0 & 2 & 3 \\ 11 & 2 & 0 & 4 \\ 5 & 3 & 4 & 0 \end{bmatrix} \end{array}.$$

Applying the single-linkage hierarchical procedure twice, the two clusters that result are

A. $\{2, 3\}$ and $\{1, 4\}$

B. $\{1, 2, 3\}$ and $\{4\}$

C. $\{1, 3, 4\}$ and $\{2\}$

D. $\{2, 3, 4\}$ and $\{1\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a set of 3 paired observations on $(x_i, b_i), i = 1, 2, 3$ as $((1, 6), (-1, 3), (3, 15))$. For the closest line b to go through these points, which of the following is the least squares solution $(\hat{\theta})$?

A. $(3, 4)$

B. $(4, 3)$

C. $(5, 3)$

D. $(3, 5)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous, smooth function whose derivative $f'(x)$ is also continuous. Suppose f has a unique global minimum $x^\star \in (-\infty, \infty)$, and you are using gradient descent to find $x^\star$. You fix some $x^{(0)} \in R$ and step size $\eta > 0$, and run $x^{(t)} = x^{(t-1)} - \eta f'\left(x^{(t-1)}\right)$ repeatedly. Which of the following statements are true?

A. Gradient descent is sure to converge, to some value, for any step size $\eta > 0$.

B. If f has a local minimum x' different from the global one, i.e., $x' \neq x^\star$, and $x^{(t)} = x'$ for some t, gradient descent will not converge to $x^\star$.

C. Assuming gradient descent converges, it converges to $x^\star$ if and only if f is convex.

D. If, additionally, f is the objective function of logistic regression, and gradient descent converges, then it converges to $x^\star$.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We use a Naive Bayes classifier to classify a single binary feature $x \in \{0, 1\}$ into two classes C_1 and C_2.

Assume that

- $P(y = C_1) = P(y = C_2) = \frac{1}{2}$
- $P(x = 1 \mid y = C_1) = p_1$
- $P(x = 1 \mid y = C_2) = p_2$

Which of the following correctly represents $\ln \dfrac{P(y = C_1 \mid x)}{P(y = C_2 \mid x)}$?

A. $x \ln \dfrac{p_1}{p_2} + (1 - x) \ln \dfrac{1 - p_1}{1 - p_2}$

B. $\ln \dfrac{p_1 + p_2}{2}$

C. $\dfrac{p_1}{p_2}$

D. $x(p_1 - p_2)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the k-means clustering algorithm applied to a dataset consisting of one-dimensional points, i.e., all data points lie on the real line and belong to $\mathbb{R}$. Assume that the number of clusters is $k = 2$, and the distance measure used is the squared Euclidean distance.

A clustering configuration is called a fixed point of the k-means algorithm if, after one complete iteration consisting of (i) assigning each point to its nearest cluster centroid and (ii) recomputing the centroids as the mean of the assigned points, both the cluster assignments and the centroids remain unchanged.

Which of the following datasets of one-dimensional points has the property that k-means can converge to two different fixed points depending on the initial cluster assignments?

A. $\{0, 1\}$

B. $\{-1, 0, 1\}$

C. $\{0, 2, 4\}$

D. $\{-2, 0, 2\}$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a binary classification problem with N training samples

$$(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N),$$

where for each $i = 1, 2, \ldots, N$, $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$.

A linear classifier with parameter vector $\mathbf{w} \in \mathbb{R}^d$ is trained using the exponential loss function defined as

$$L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^{N} \exp\left(-y_i \, x_i^\top \mathbf{w}\right) .$$

Which of the following statements is *true*?

A. The loss $L(\mathbf{w})$ increases when training samples are badly misclassified.

B. There exists a vector $\mathbf{w}^\star$ such that $L(\mathbf{w}^\star) = 0$.

C. If there exists a vector $\mathbf{w}^\star$ such that $L(\mathbf{w}^\star) < \frac{1}{N}$, then $\mathbf{w}^\star$ correctly classifies all training samples.

D. If all training samples are correctly classified, then $L(\mathbf{w}) < 1$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider **randomized** QuickSort and fix an element t.

Let X_{i-1} be the size of the subproblem containing a fixed element t at recursion level $i-1$, and let X_i be the size of the subproblem containing t at the next recursion level. Let $X_0 = n$.

A pivot is called **good** if it lies in the middle of the subarray. More specifically, the pivot is not among the smallest 25% or the largest 25% elements of the subarray.

A pivot is called **bad** if, even at the next recursion level, the size of the subproblem containing t remains X_{i-1} .

Assume that a pivot is good with probability $\frac{1}{2}$.

The quantity $\mathbb{E}[X_i \mid X_{i-1}]$ denotes the expected size of the subproblem containing t at the next recursion level, given that the current subproblem size is X_{i-1}.

Which of the following is a correct upper bound on $\mathbb{E}[X_i \mid X_{i-1}]$?

A. $\frac{3}{4} X_{i-1}$

B. $\frac{7}{8} X_{i-1}$

C. $\frac{1}{2} X_{i-1}$

D. X_{i-1}

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

A binary counter is a sequence of k bits $b_{k-1} b_{k-2} \ldots b_0$, where each bit is either 0 or 1. The operation INCREMENT increases the value of the counter by 1 (in base 2).

The cost of one INCREMENT is the number of bits whose value changes during that INCREMENT.

Assume $k = 4$. Starting from all bits equal to 0, the first $n = 6$ INCREMENT operations produce the following values:

INCREMENT Number	b_3	b_2	b_1	b_0
–	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0

For example, consider INCREMENT number 4. The counter changes from 0011 to 0100. Here, three bits change their values, so the cost of INCREMENT number 4 is 3.

Let $\mathrm{WCSC}(n)$ denote the total cost of performing n INCREMENT operations starting from all 0's.

Which of the following is correct?

A. $\mathrm{WCSC}(n) = \Theta(n \log_2 n)$

B. $\mathrm{WCSC}(n) = \Theta(n)$

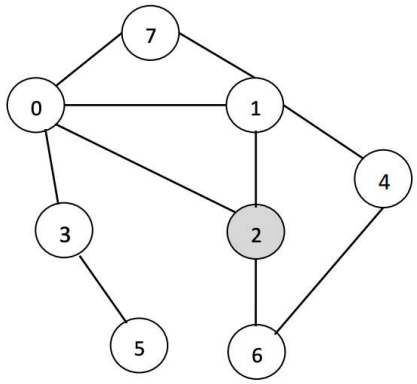
C. $\mathrm{WCSC}(n) = \Theta(2^k)$

D. $\mathrm{WCSC}(n) = \Theta(nk)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the undirected graph shown below.



A Breadth-First Search (BFS) traversal is performed starting from vertex 2. Whenever the neighbors of a vertex are examined, they are processed in increasing order of vertex number.

Let $d(v)$ denote the distance (number of edges) from the source vertex 2 to vertex v in the BFS tree constructed.

Determine the value of $d(5) + d(7)$.

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Letters are pushed onto a stack in the following order:

$$R\ A\ N\ D\ O\ M\ O\ P\ S$$

You may insert pop operations, denoted by $*$, at any positions among the pushes. Each pop outputs the current top of the stack to the output sequence.

The process is performed only once (i.e., the output cannot be passed again through the stack).

Which of the following sequences of push and pop operations correctly produces the output:

$$A\ D\ O\ N\ O\ M\ S\ P\ R$$

A. `R A * N D * O * M O * P S * * * *`

B. `R A * N D O * * M O * P S * * *`

C. `R A * N D * O * * M O * * P S * * *`

D. It is not possible to generate the output string `ADONOMSPR` using a single stack pass.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

A patient goes to the doctor for a medical condition, and the doctor suspects 3 diseases as the cause of the condition. The 3 diseases are D_1, D_2, and D_3, and they are independent from each other (given no other observations). There are 4 symptoms S_1, S_2, S_3, and S_4, and the doctor wants to check for their presence in order to find the most probable cause. Suppose we know that

- S_1 can be caused by D_1
- S_2 can be caused by D_1 and D_2
- S_3 can be caused by D_1 and D_3
- S_4 can be caused by D_3.

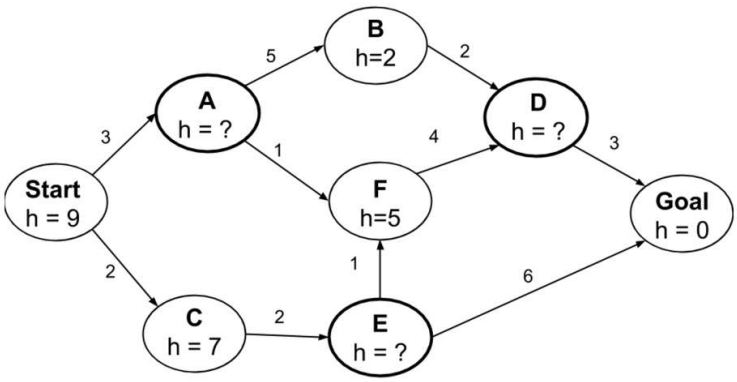
Assume all random variables are Bernoulli, i.e. the patient has the disease/symptom or not.

How many parameters are required to describe this joint distribution?

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the state-space graph shown above. Some states are missing heuristic values. Each edge is labeled with its step cost. The heuristic values for some states are given.



The heuristic function $h(\cdot)$ is required to be both *admissible* and *consistent*.

Which of the following options correctly specifies the possible range(s) of the missing heuristic values?

(More than one option may be correct.)

A. $6 \leq h(A) \leq 6$

B. $1 \leq h(D) \leq 3$

C. $5 \leq h(E) \leq 6$

D. $0 \leq h(D) \leq 2$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Recall that during Gibbs Sampling, samples are generated through an iterative process. Assume that the only evidence that is available is $A = +a$. Clearly fill in the circle(s) of the sequence(s) below that could have been generated by Gibbs Sampling.

A. Sequence 1

	a	b	c	d
1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$+d$
3	$+a$	$-b$	$+c$	$+d$

B. Sequence 2

	a	b	c	d
1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$-d$
3	$-a$	$-b$	$-c$	$+d$

C. Sequence 3

	a	b	c	d
1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$-d$
3	$+a$	$+b$	$-c$	$-d$

D. Sequence 4

	a	b	c	d
1	$+a$	$-b$	$-c$	$+d$
2	$+a$	$-b$	$-c$	$-d$
3	$+a$	$+b$	$-c$	$+d$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the relations $\mathbf{R}(\underline{\mathrm{A}}, \mathrm{B}*)$ and $\mathbf{S}(\mathrm{A}*, \underline{\mathrm{B}})$, where $\mathbf{R}$ has a foreign key referencing $\mathbf{S}$ via B , and $\mathbf{S}$ has a foreign key referencing $\mathbf{R}$ via A.

Which of the following is guaranteed to produce fewer than, or at most the same, number of tuples as any of the others?

A. $\mathbf{R} \bowtie \mathbf{S}$

B. $\mathbf{R} \cup \mathbf{S}$

C. $\mathbf{R} \bowtie \pi_{\mathrm{B}}(\mathbf{S})$

D. $\pi_{\mathrm{A}}(\mathbf{R}) \bowtie \mathbf{S}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

The relation $\mathrm{R(a, b)}$ may have duplicate tuples. Which of the following SQL queries has a result that is guaranteed not to have duplicates, regardless of what tuples R contains?

A. $\mathrm{SELECT\ a\ FROM\ R\ WHERE\ a} = 1$

B. $\mathrm{SELECT\ MAX(b)\ FROM\ R\ GROUP\ BY\ a}$

C. $\mathrm{SELECT\ a, b\ FROM\ R\ GROUP\ BY\ a, b}$

D. $\mathrm{SELECT\ a\ FROM\ R\ WHERE\ a\ NOT\ IN\ (SELECT\ a\ FROM\ R)}$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following database schema:

$$\mathbf{R}(\underline{\mathrm{A}}, B)\ \mathrm{FK}\ (\mathrm{B})\ \mathrm{refs}\ \mathbf{R}(\mathrm{A})$$

R has a foreign key on B that references $\mathrm{R(A)}$. A is the primary key of R.

What is the largest that $|\mathbf{R} \bowtie \pi_{\mathrm{A}\to\mathrm{B},\mathrm{B}\to\mathrm{A}}(\mathbf{R})|$ can be?

A. 0

B. $|\mathbf{R}|$

C. $2|\mathbf{R}|$

D. $|\mathbf{R}|^2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $P(x)$ and $Q(x)$ be predicates and let D denote the domain of the predicate variable x. Consider the following universal conditional statement,

$$\forall x \in D, P(x) \to Q(x).$$

Which of the following conditions implies that the above universal conditional statement is true?

A. $P(x) \wedge Q(x)$ is false for all $x \in D$.

B. $P(x) \wedge (\sim Q(x))$ is false for all $x \in D$.

C. $Q(x)$ is true for all $x \in D$.

D. $P(x) \vee Q(x)$ is true for all $x \in D$.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

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